

Advanced Portfolio with CSS Grid

Week 5 — Project Documentation

Arathi Shekhar Munavalli | The Developers Arena Internship

Live Demo: https://arathism.github.io/week5_Portfolio/

GitHub Repository: https://github.com/arathism/week5_Portfolio

1. Project Overview & Objectives

This project is a rebuild of my personal portfolio (originally created in Week 3) using advanced CSS techniques. The objective was to move from a flatter, less structured stylesheet to a fully systematic layout using CSS Grid, Flexbox, CSS custom properties (variables), keyframe animations, advanced selectors, and consistent BEM naming — while keeping all original interactive functionality intact and adding a new feature.

The page retains its three original interactive features and upgrades the CSS/JS behind each one:

- Dark / light mode toggle, with the choice saved in localStorage so it persists across reloads.
- A to-do list supporting add, mark-done, and remove actions.
- A contact form with real-time validation and inline error/success states.

A new Projects tile was also added, showcasing past coursework — including the Week 4 "SecureStack" business website — as a CSS Grid card layout with hover interactions.

2. Setup and Installation Instructions

No build tools, package managers, or dependencies are required — this is a plain HTML/CSS/JavaScript project.

Option A — Open directly

Download or clone the repository, then open index.html directly in any modern browser.

Option B — Serve locally (recommended)

```
git clone https://github.com/arathism/week5_Portfolio.git
cd week5_Portfolio
python3 -m http.server 8000
# then visit http://localhost:8000
```

Trying the interactive features

- Click the 🌙 / ☀️ icon in the navbar to switch themes (persists on reload).
- Add a task in the To-Do tile; click its text to mark done; click "Remove" to delete it.
- Hover over a Projects card to see the lift/shadow animation and tag hover states.
- Submit the Contact form empty to see validation errors, then fill it in to see errors clear in real time and a success message appear.

- Resize the browser (or open on a phone) below 700px to see the collapsible hamburger nav and single-column grid.

3. Code Structure Explanation

index.html	Page markup – navbar, hero, tile grid, footer
css/	
└─ main.css	CSS variables (theme tokens), reset, typography, buttons
└─ layout.css	CSS Grid + Flexbox structure, mobile-first breakpoints
└─ animations.css	Keyframes, transitions, hover/focus, pseudo-elements
js/	
└─ theme-switcher.js	Theme toggle, mobile nav, to-do list, form validation
images/	
└─ profile.jpg	About tile photo
projects/securestack/	Week 4 site kept as a local reference copy
screenshots/	Visual documentation

Total project size: 1,122 lines across index.html (185), layout.css (411), animations.css (230), theme-switcher.js (168), and main.css (128).

4. Screenshots of Working Application

The screenshots below demonstrate the site running at desktop and mobile breakpoints, in both themes, and with each interactive feature exercised.

4.1 Desktop — Light Theme (1280×900)

SM_A

AboutSkillsProjectsTo-DoContact

Hi, I'm Arathi Shekhar Munavalli

Final-Year CSE Student · Open to Internships

About Me

Arathi Shekhar Munavalli

I'm a final-year Computer Science and Engineering student who loves the moment a stubborn bug finally gives up and the code just works. My focus these days is full-stack development, AI-powered applications, and cybersecurity — I like understanding not just how to build something, but how someone might try to break it, and building it to hold up anyway. Outside of coursework I've picked up hands-on experience through internships and hackathons, and I'm always looking for the next problem worth solving. Take a look at the Projects tile below for what that's turned into so far — and I'm currently open to internship and full-time opportunities where I can keep learning by building.

Skills

HTML / CSS

JavaScript

React

Node.js

Python

MongoDB

Quick To-Do List

Add a task...

Add

No tasks yet — add one above.

Projects

SecureStack

A multi-page business website for a fictional web-development & cybersecurity consulting studio. Built with semantic HTML, a mobile-first Grid/Flexbox layout, and vanilla JS form validation using the Constraint Validation API. Includes Home, Services, About, and Contact pages with a shared design system.

HTML5

CSS Grid

Flexbox

Vanilla JS

Accessibility

View Project

Source Code

This Portfolio

The site you're looking at — a personal portfolio rebuilt with CSS Grid for the main layout, Flexbox for components, CSS custom properties for a light/dark theme system, keyframe animations, and BEM naming throughout.

CSS Grid

CSS Variables

BEM

Animations

View Project

Source Code

Contact Me

Name

Email

Message

Send Message

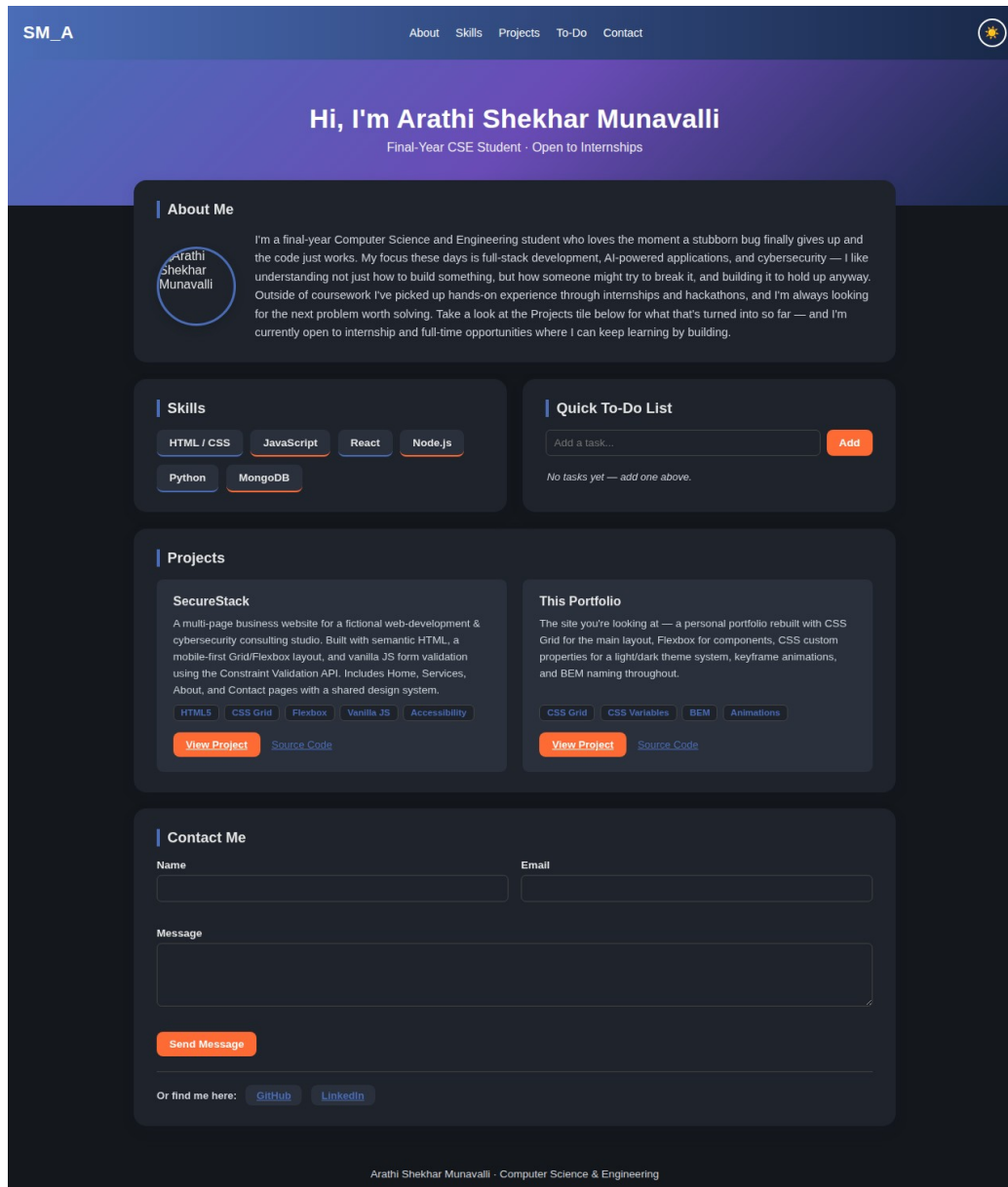
Or find me here:

GitHub

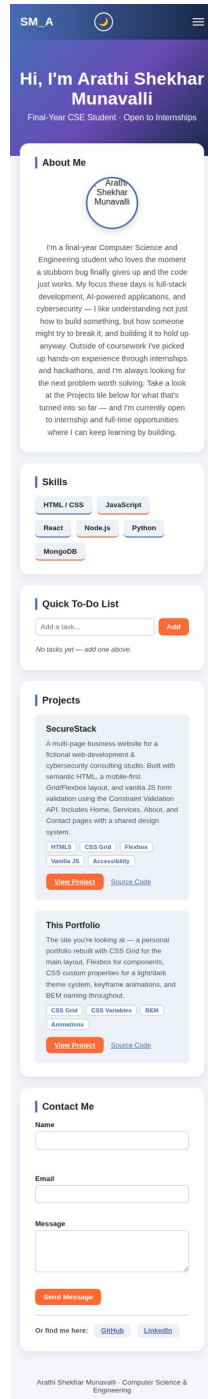
LinkedIn

Arathi Shekhar Munavalli · Computer Science & Engineering

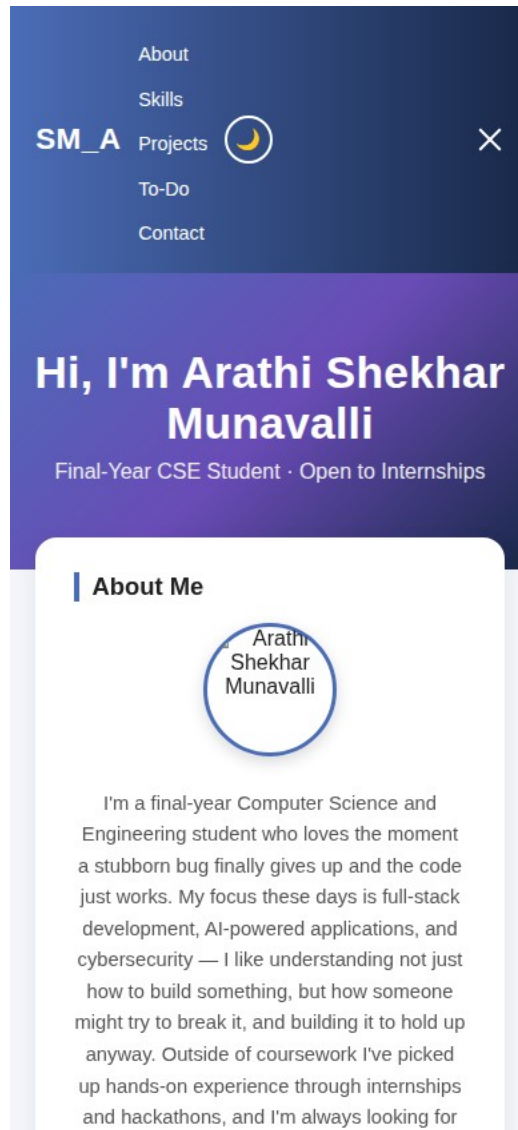
4.2 Desktop — Dark Theme (1280×900)



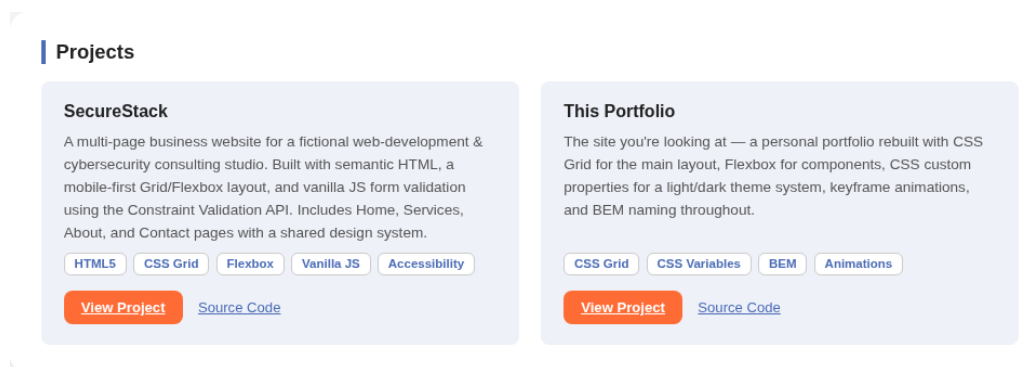
4.3 Mobile View (390×844)



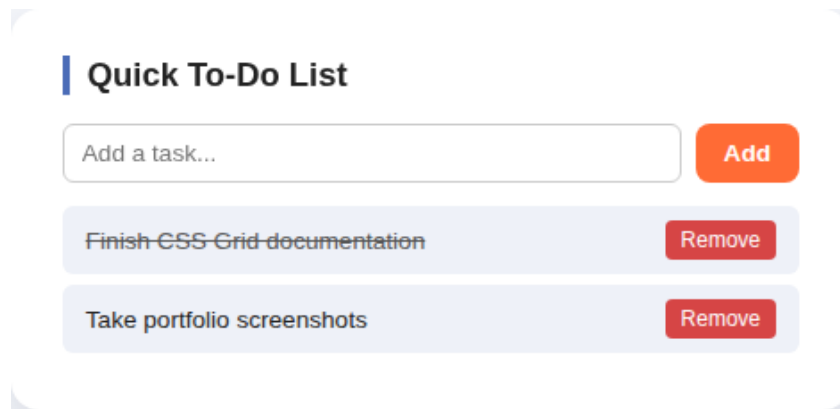
4.4 Mobile Navigation Open



4.5 Projects Section



4.6 To-Do List



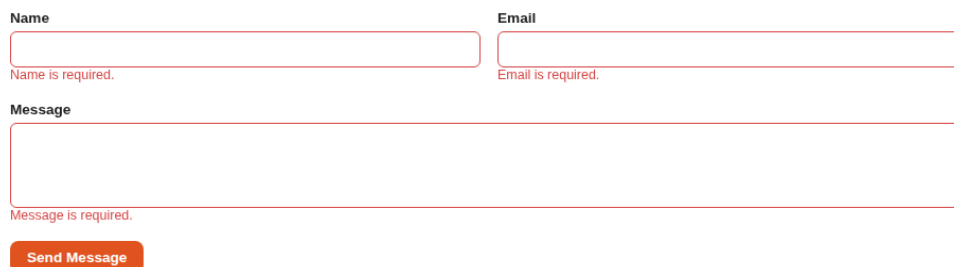
Quick To-Do List

Add a task... **Add**

Finish CSS Grid documentation **Remove**

Take portfolio screenshots **Remove**

4.7 Contact Form — Validation Error State



Name

Email

Name is required.

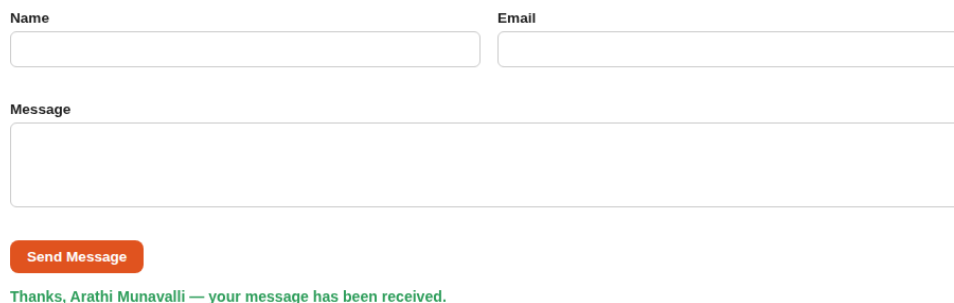
Email is required.

Message

Message is required.

Send Message

4.8 Contact Form — Success State



Name

Email

Message

Send Message

Thanks, Arathi Munavalli — your message has been received.

5. How Technical Requirements Were Met

5.1 CSS Grid (main layout)

- The main `.tile-grid` uses `display: grid` with `grid-template-columns: repeat(2, 1fr)`; the About and Contact tiles span both columns while Skills and To-Do sit side by side — explicit grid placement, not just auto-flow.
- `.projects` is its own 2-column CSS Grid of `.project-card` items, collapsing to 1 column under 700px — a second, nested example of the Grid system inside a wide tile.
- The contact form's Name/Email row (`.contact-form__row`) is its own 2-column grid, also collapsing to 1 column under 700px.
- Below 700px, `.tile-grid` collapses to a single column and wide tiles drop their span — handled with mobile-first min/max-width media queries.

5.2 Flexbox (components)

- `.navbar` uses Flexbox (justify-content: space-between) to lay out the logo, hamburger toggle, link list, and theme button, switching `.navbar__links` from a collapsible column (mobile) to a horizontal row ($\geq 700\text{px}$) via flex-direction.
- `.about` and `.skills` use Flexbox for the avatar/text pairing and wrapping skill tags respectively.
- `.project-card__body` uses flex-direction: column with the description set to flex-grow: 1, so card footers align to the bottom evenly across cards of different description lengths.
- `.todo__input-row` and `.todo__item` use Flexbox for the input+button row and the space-between layout of each task.

5.3 CSS Variables (theme system)

- All colors, gradients, spacing, radii, and easing curves are defined once in `:root` in `main.css`.
- `[data-theme="dark"]` re-declares the same variable names with dark values — no duplicated component rules, only the tokens change. The Projects tile reuses the same tokens (`--color-surface-alt`, `--color-primary`, etc.), so it themes automatically with no extra rules.
- `theme-switcher.js` toggles the `data-theme` attribute on `<html>` and persists the choice in `localStorage`, falling back to the OS-level `prefers-color-scheme` on first visit.

5.4 Animations & Transitions

- `@keyframes fadeInUp` staggers each tile in on page load using animation-delay per `nth-child`.
- `@keyframes popIn` animates new to-do items in as they're added.
- `@keyframes shake` plays on invalid form fields to draw attention without relying on color alone.
- Hover states (tiles, skill boxes, buttons, project cards, project tags) animate transform, box-shadow, background, and color — properties that stay off the layout/paint-heavy path for smoother performance.
- **Performance note:** `prefers-reduced-motion: reduce` is respected globally — all animation and transition durations collapse to near-zero for users who've requested reduced motion.

5.5 Advanced Selectors & Pseudo-elements

- `.navbar__link::after` draws an animated underline on hover/focus instead of a static border; `.project-card__link--ghost::after` reuses the same pattern for "Source Code" links.
- `.todo__list:empty::before` shows a "No tasks yet" placeholder using the `:empty` selector — no JavaScript needed for that piece of UI state.
- `.skills__box:nth-child(odd/even)` alternates an accent underline between primary and accent colors.
- `:focus-visible` (not plain `:focus`) is used on all interactive elements, so keyboard users get a visible outline without a ring appearing on every mouse click.
- Attribute selectors (`[data-theme="dark"]`, `[aria-expanded="true"]`) drive theming and the hamburger-to-X icon animation.

5.6 BEM Methodology

Every class follows `block__element--modifier`, e.g. `.tile-grid__tile--wide`, `.todo__item--done`, `.navbar__links--open`, `.contact-form__input`, `.project-card__link--ghost`. This was the main structural change from the Week 3

version, which used flatter, less consistent names (.tile, .skill-box, .todo-list li) — this pass renames everything into one consistent BEM system, including the newly added .project-card block.

5.7 Layout Decisions

- Mobile-first breakpoints (single 700px breakpoint) keep the CSS lean rather than stacking multiple overlapping media queries.
- Grid was chosen for 2-dimensional layouts (the tile grid, the project cards, the form row); Flexbox was chosen for 1-dimensional component-level layout (navbar, card internals, list items) — using each tool for the layout problem it fits best rather than one exclusively.
- A single spacing scale (--space-xs through --space-xl) and two radius tokens keep visual rhythm consistent across every tile without repeated magic numbers.

5.8 Performance Optimizations

- Animations and transitions are restricted to transform, opacity, box-shadow, background, and color — properties that avoid layout thrashing (no animating width/height/top/left directly where a transform equivalent exists).
- Theme switching only changes CSS custom property values on the root element rather than swapping stylesheets or classes on every component, so the repaint cost is minimal.
- No external CSS/JS frameworks are loaded — plain CSS and vanilla JS keep the page lightweight with zero render-blocking third-party requests.
- prefers-reduced-motion support avoids unnecessary animation work for users who've opted out at the OS level.

6. Testing Evidence

Verified with a Chromium browser at two viewports, with zero console errors at either:

- **1280×900 (desktop):** confirmed the 2-column Grid renders with correct tile spans, the theme toggle swaps all tokens instantly with no flash of unstyled content, to-do add/remove/mark-done all work, the Projects grid shows two cards side by side with working hover states, and form validation correctly blocks submission with errors, then clears and shows a success message once all fields are valid.
- **390×844 (mobile):** confirmed the nav collapses to a hamburger, expands with a smooth height transition, animates into an "X", the tile grid stacks to one column with wide tiles no longer spanning, and the Projects grid drops to a single column.
- **Keyboard / accessibility:** tabbed through nav links, buttons, project card links, and form fields — confirmed a visible :focus-visible outline on each, and that aria-expanded / aria-label are set on the nav toggle and mode button.
- **Reduced motion:** simulated prefers-reduced-motion: reduce — all animations and transitions become effectively instant.

8. README.md (For GitHub Repository)

The following is the full README content for the project repository, including live links, screenshots, and technical documentation formatted for GitHub.

Week 5 — Advanced Portfolio with CSS Grid

Live Demo: https://arathism.github.io/week5_Portfolio/

Repository: https://github.com/arathism/week5_Portfolio

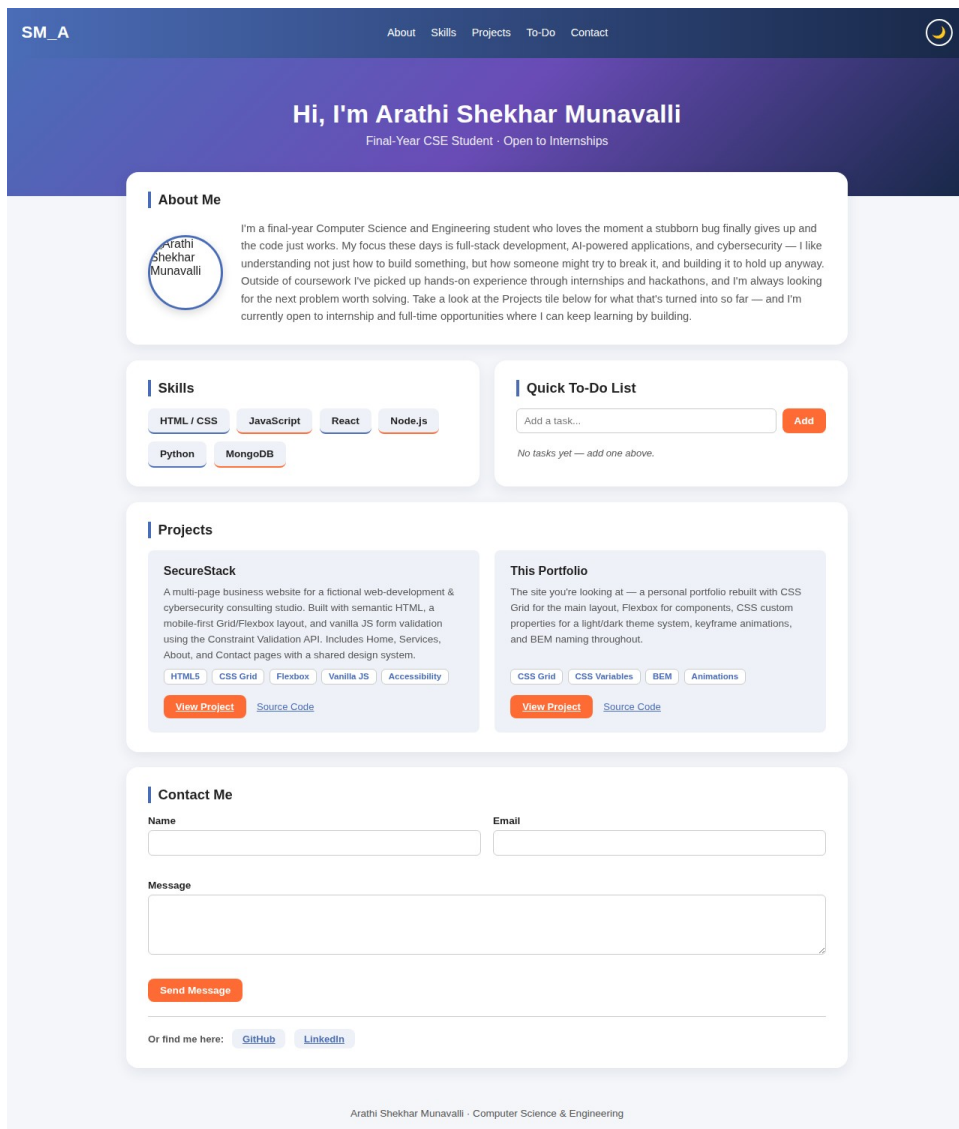
Project Overview

This is Arathi Shekhar Munavalli's personal portfolio (originally built in Week 3), rebuilt for Week 5 using advanced CSS techniques: CSS Grid for the main layout, Flexbox for components, CSS custom properties for a full light/dark theme system, keyframe animations, advanced selectors and pseudo-elements, and BEM naming throughout the stylesheet.

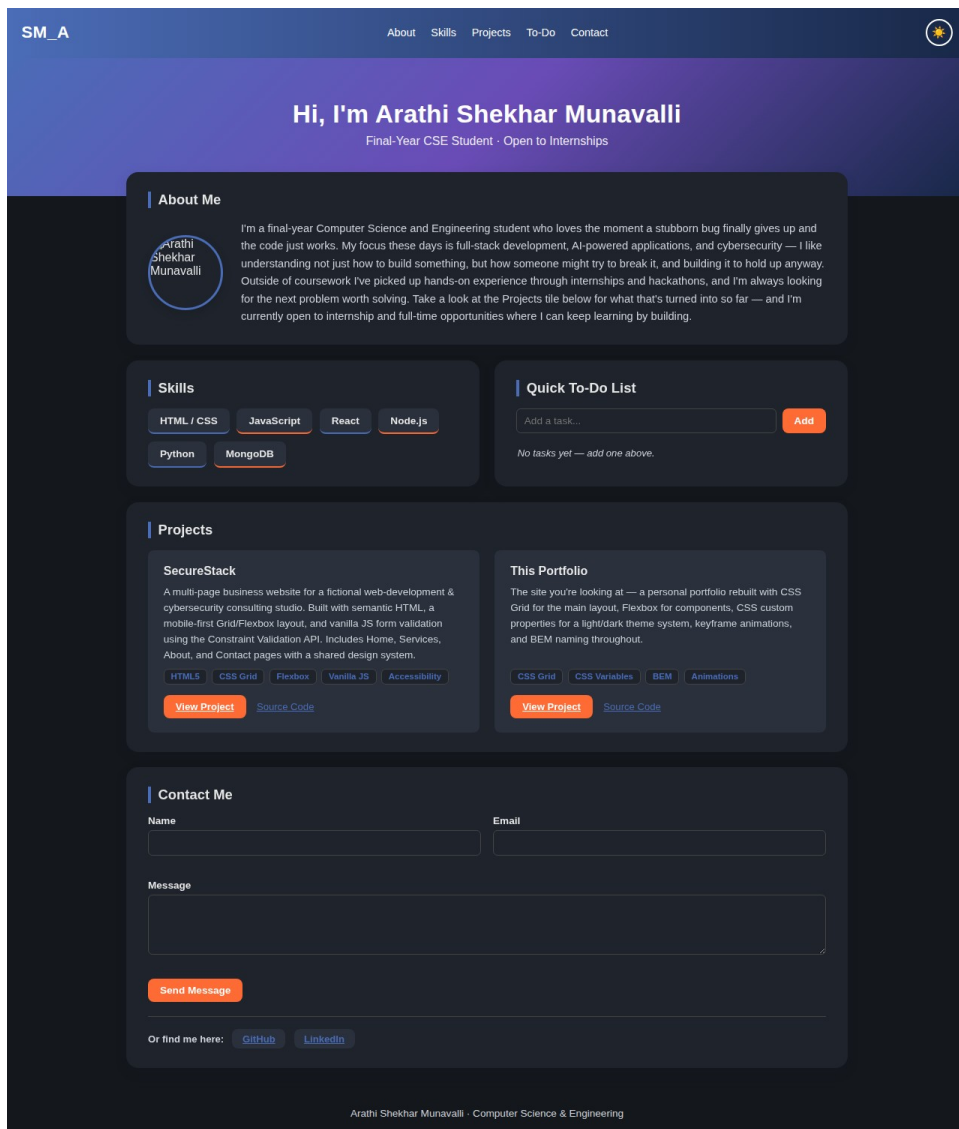
The page keeps its three original interactive features — dark/light mode toggle, a to-do list with add/remove, and real-time contact form validation — and upgrades the CSS and JS behind all three to meet this week's requirements. It also adds a new Projects tile that showcases past coursework, including the Week 4 "SecureStack" business website, as a CSS Grid card layout.

Screenshots

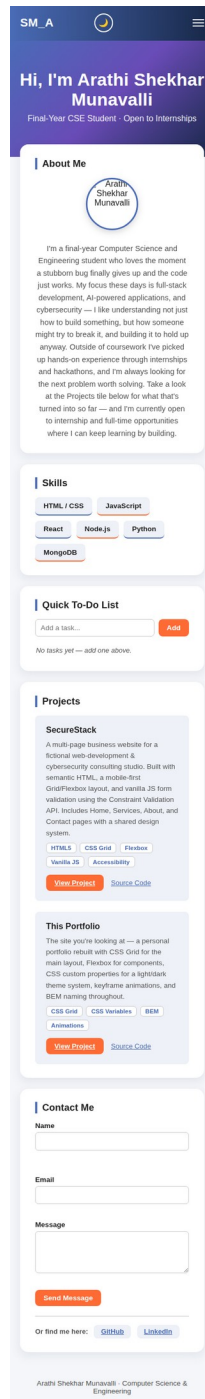
Light Theme:



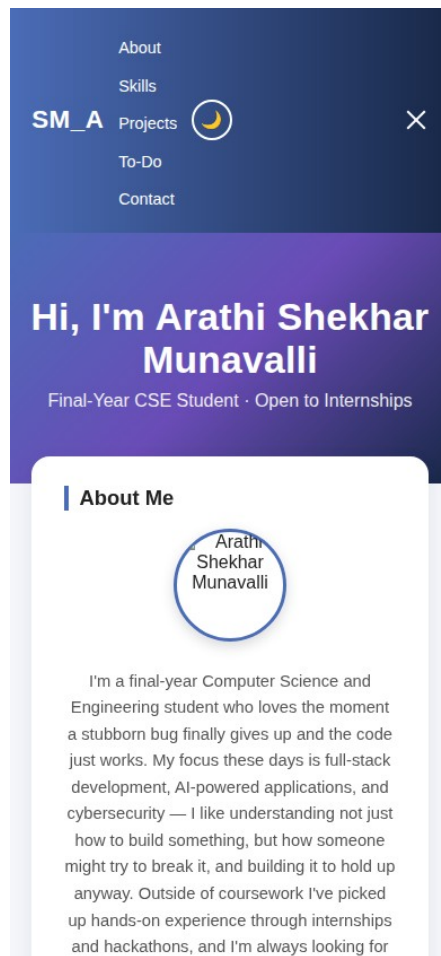
Dark Theme:



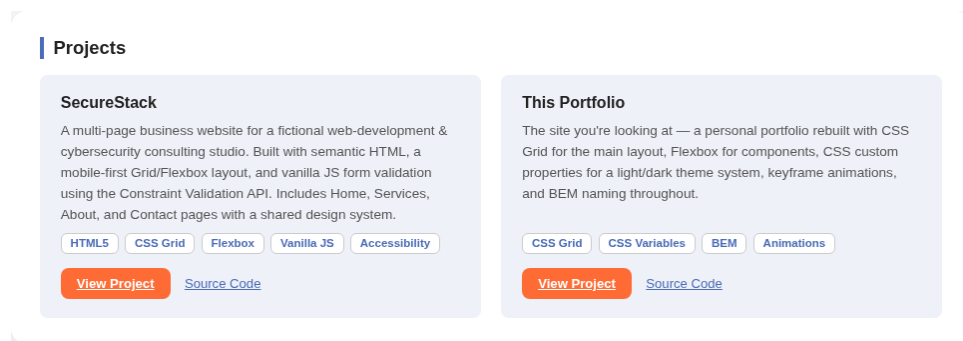
Mobile View:



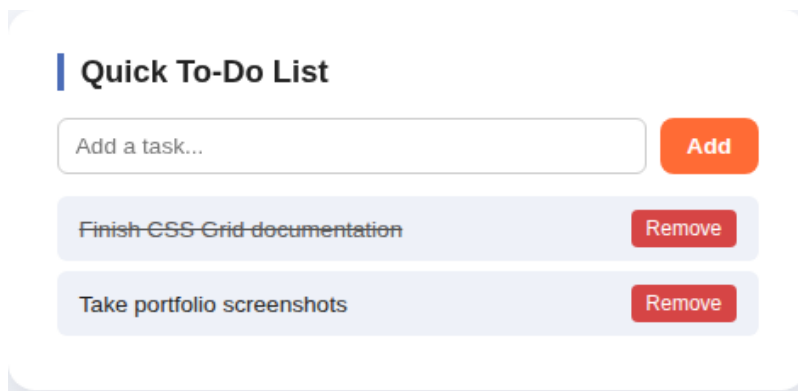
Mobile Navigation Open:



Projects Section:



To-Do List:



Contact Form — Validation Error:

Name Email

Name is required. Email is required.

Message

Message is required.

Contact Form — Success:

Name Email

Message

Thanks, Arathi Munavalli — your message has been received.

Setup Instructions

- Download or clone this folder.
- No build tools or dependencies are needed — plain HTML/CSS/JS.
- **Open index.html directly** in a browser, or serve it locally (recommended):

```
python3 -m http.server 8000
# then visit http://localhost:8000
```

Try the interactive features:

- Click the 🌙/☀️ icon in the nav to switch themes (choice is saved in localStorage, so it persists on reload).
- Add a task in the To-Do tile, click its text to mark it done, click "Remove" to delete it.
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- Resize the browser (or open on a phone) below 700px to see the collapsible hamburger nav and single-column grid.

Code Structure

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js/	
└─ theme-switcher.js	Theme toggle, mobile nav, to-do list, form validation
images/	
└─ profile.jpg	About tile photo
projects/securestack/	Week 4 site kept as a local reference copy
screenshots/	Visual documentation
README.md	

CSS is split by concern: main.css owns what things look like (tokens, type, buttons), layout.css owns where things sit (Grid/Flexbox, breakpoints), animations.css owns how things move and react. Keeping these separate avoids specificity clashes between layout rules and animation/hover rules targeting the same elements.

Notes

- No CSS minification/build step is included per assignment scope.

9. Project Links

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GitHub Repository: https://github.com/arathism/week5_Portfolio